



SeriScope: A Low-Cost Edge-AI System for Fertility and Disease Assessment of Tassar Silkworm Eggs

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Abstract

Pebrine disease, which is caused by *Nosema bombycis*, is one of the most harmful threats to the sericulture sector Govindan et al. (1997). Traditional diagnosis involves manual microscopic analysis of moth body fluid, which is **time-consuming** and also liable to **human errors** Meijering et al. (2016), Hallou et al. (2021). This paper describes a **low-price Edge AI diagnosis tool named SeriScope** that automatically diagnoses pebrine disease and calculates egg fertility through microscopic image analysis Litjens et al. (2017), Shi et al. (2016), Satyanarayanan (2017). This tool makes use of five different classification models and has been trained on more than **1400 real-world field dataset** examples collected from the tassar grainages across the regions affected by sericulture. Operating fully under an offline environment on embedded hardware platforms, SeriScope can provide fast diagnosis with least infrastructure Shi et al. (2016), Satyanarayanan (2017) and thereby assist farmers while supporting rural economic growth aligned with SDG Goals 8 and 9 United Nations (2015).

KEYWORDS

Pebrine, Time-consuming, Human errors, SeriScope

1 INTRODUCTION

Animal science is a broad field that combines biology, physiology, genetics, pathology, microbiology, ecology, nutrition, and technology Quinn et al. (2011). It aims to understand, manage, and improve the health and productivity of animals. Since ancient times, animals have been vital for human survival, providing food, clothing, work, transportation, and economic security. Today, animal-based industries are crucial parts of rural economies, especially in developing countries where agriculture and related sectors offer most jobs and income. The health and well-being of animals directly affect productivity, sustainability, and economic stability, making early disease diagnosis a key concern World Organisation for Animal Health (2024), Quinn et al. (2011).

Across animal production systems (livestock, poultry, aquaculture, sericulture, and wildlife management), infectious disease outbreaks remain a persistent challenge World Organisation for Animal Health (2024), Quinn et al. (2011). Infectious diseases caused by bacteria, viruses, protozoa, fungi, and parasites can spread rapidly and lead to reduced productivity and economic loss World Organisation for Animal Health (2024). Many systems fail to detect diseases early due to subtle symptoms, lack of diagnostic tools, or reliance on expert interpretation World Organisation for Animal Health (2024). Consequently, by



FIGURE 1 Field visit to Sukinda Silk Farm with a sericulture expert, showing the tassars host plants (e.g., *Terminalia arjuna* and *Terminalia tomentosa*/Asana) used in tassars rearing.

the time visible symptoms appear, damage may be difficult to reverse Quinn et al. (2011).



FIGURE 2 Traditional microscopy-based diagnosis workflow used in grainages: manual observation of smears for pathogen identification.

Traditional animal health diagnosis relies on visual inspection, clinical symptoms, lab tests, and microscopic examinations World Organisation for Animal Health (2024). Microscopy remains one of the most trusted diagnostic tools for direct visualization of pathogens and abnormalities Meijering et al. (2016), Hallou et al. (2021). However, microscopic diagnosis depends on skilled personnel, consistent sample preparation, and subjective interpretation, which can be limiting in rural settings Hallou et al. (2021), Litjens et al. (2017).

Sericulture, the rearing of silkworms for silk production, is a delicate animal-production system with short life cycles and high sensitivity to environmental and biological stressors Hasan et al. (2025). Tassar sericulture is ecologically and economically important and is often practiced in forested/semi-forested regions by marginal communities. Despite its significance, sericulture faces serious disease threats, with pebrine among the most destructive Govindan et al. (1997). Pebrine is caused by *Nosema bombycis*, and vertical transmission through eggs makes it particularly hazardous Govindan et al. (1997). Traditional pebrine detection involves identifying spores under a microscope, requiring expertise and time Govindan et al. (1997), Meijering et al. (2016). In many grainages, limited access to skilled microscopists and facilities increases the chance of infected eggs reaching farmers Govindan et al. (1997).

Similar field-diagnostic constraints exist across animal science, motivating rapid, affordable, and scalable tools World Organisation for Animal Health (2024). In recent decades, digital imaging, computational tools, and artificial intelligence have enabled new approaches for biological diagnostics Litjens et al. (2017), Hallou et al. (2021). In image-based diagnostics, deep learning—particularly Convolutional Neural Networks (CNNs)—learns discriminative visual features directly from pixel data and has transformed medical imaging and bioimage analysis Litjens et al. (2017). However, many reported AI diagnostic pipelines require high-performance computing or cloud processing, which may be impractical in rural environments due to connectivity and cost constraints Satyanarayanan (2017), Shi et al. (2016).



FIGURE 3 Field sample collection and microscopy setup used for slide preparation and image capture under grainage conditions.

Edge computing offers a practical alternative: Edge AI deploys inference directly on local embedded devices, enabling offline operation, lower latency, and improved privacy Shi et al. (2016), Satyanarayanan (2017). For field diagnostics, this enables real-time decisions at farms and grainages. Combining edge AI with microscopy can broaden access to objective diagnostics and reduce dependence on centralized labs Litjens et al. (2017), Shi et al. (2016). These outcomes support sustainable development goals through productivity improvements and rural innovation United Nations (2015).

This research addresses sericulture diagnostic gaps by developing a low-cost, offline, AI-powered system for early pebrine detection using deep learning on field microscopy images and embedded deployment Litjens et al. (2017), Shi et al. (2016). Beyond sericulture, the underlying workflow (microscopy imaging, preprocessing, CNN classification, edge deployment) is applicable to broader animal health diagnostics World Organisation for Animal Health (2024), Hallou et al. (2021).

2 MATERIALS AND METHODS

This section provides a detailed overview of the biological materials, experimental design, image acquisition procedures, computational framework, preprocessing pipeline, image analysis methods, and software used in developing and evaluating the proposed Edge-AI-based diagnostic system Hallou et al. (2021), Meijering et al. (2016), Shi et al. (2016). The methodology aimed to ensure scientific accuracy, reproducibility, reliability, and practicality in real-world situations Sandve et al. (2013), World Organisation for Animal Health (2024).

2.1 Experimental Setup

2.1.1 | Study Design and Overall Framework

The experimental design followed a multistage diagnostic pipeline incorporating biological sample collection, microscopic imaging, preprocessing, deep learning-based classification, and deployment testing Hallou

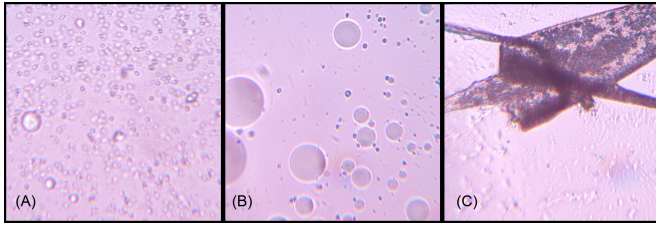


FIGURE 4 Representative microscopy images from the dataset showing (a) pebrine-positive smear, (b) non-pebrine/healthy smear, and (c) bacterial contamination.

et al. (2021), Litjens et al. (2017), Shi et al. (2016). The workflow consisted of:

1. Field-level biological sample collection World Organisation for Animal Health (2024)
2. Laboratory and on-site slide preparation Meijering et al. (2016)
3. Microscopic image acquisition Meijering et al. (2016), Hallou et al. (2021)
4. Image preprocessing and normalization Gonzalez and Woods (2018)
5. Deep learning-based image analysis Litjens et al. (2017)
6. Model evaluation using statistical metrics Powers (2020)
7. Edge deployment and performance assessment Shi et al. (2016), Satyanarayanan (2017).

2.1.2 | Biological Material and Sample Source

Biological samples were collected from tassard silkworm (*Antheraea mylitta*) populations maintained in grainages and sericulture clusters Hasan et al. (2025). Pebrine infection (spores of *Nosema bombycis*) and related diagnostic considerations follow established sericulture pathology knowledge Govindan et al. (1997).

2.1.3 | Ethical and Biosafety Considerations

All biological handling followed standard biosafety and diagnostic hygiene practices to prevent cross-contamination World Organisation for Animal Health (2024). Infected samples were isolated immediately after identification consistent with containment guidelines for infectious agents World Organisation for Animal Health (2024).

2.1.4 | Microscopy Infrastructure

Microscopic imaging used bright-field compound microscopes typical of field and laboratory diagnostic settings Meijering et al. (2016). Image capture using digital/USB microscope cameras is consistent with standard bioimage analysis practice Hallou et al. (2021).

2.1.5 | Image Acquisition Protocol

Imaging protocols using multiple fields of view and consistent illumination are standard methods to reduce sampling bias and improve representativeness Meijering et al. (2016), Hallou et al. (2021).

2.1.6 | Experimental Hardware for Edge Deployment

Edge testing on embedded platforms without dedicated GPUs aligns with typical constraints in edge AI deployments Shi et al. (2016), Satyanarayanan (2017).

2.2 Software Implementation

2.2.1 | Computational Environment

All computational experiments were performed using open-source frameworks to ensure transparency and reproducibility Sandve et al. (2013). The implementation was developed primarily in **Python 3** Python Software Foundation (2022) using **PyTorch** as the core deep-learning framework Paszke et al. (2019), alongside community-standard scientific libraries. The software stack included:

- **Deep learning and model training:** torch, torchvision Paszke et al. (2019)
- **Dataset handling:** ImageFolder, DataLoader, random_split main-tainers and contributors (2016)
- **Hyperparameter optimization:** Optuna Akiba et al. (2019)
- **Metrics and reporting:** scikit-learn Pedregosa et al. (2011)
- **Image preprocessing utilities:** PIL, torchvision.transforms Murray et al. (2026)
- **Edge runtime benchmarking:** ONNX Runtime ONNX Runtime developers (2018)
- **Real-time inference and camera interfacing:** OpenCV OpenCV developers (2026)
- **Model export and optimization:** ONNX, TorchScript, mobile optimization ONNX developers (2026), Paszke et al. (2019)

Model training and validation were performed on a workstation with automatic device selection (CUDA where available, otherwise CPU). Inference benchmarking and deployment-style evaluation were conducted using CPU runtimes to emulate embedded field deployment conditions.

2.2.2 | Dataset Organization and Annotation

The dataset was organized into structured directories following a class-wise folder hierarchy compatible with torchvision.datasets.ImageFolder Paszke et al. (2019). Class labels were inferred directly from directory names to ensure consistency between training and evaluation.

Annotation followed both binary and extended multi-class labeling strategies depending on the experimental configuration. All labels were verified through microscopic confirmation, consistent with best practices in biological image analysis Hallou et al. (2021).

2.2.3 | Training-Validation Strategy

An **80:20** training-validation split was implemented using stratified random sampling to ensure balanced class representation Kohavi (1995). Validation data were used exclusively for model selection and checkpointing.

Independent field-acquired samples were reserved for testing to assess real-world generalization, a recommended practice for avoiding dataset leakage and site bias Torralba and Efros (2011).

3 Preprocessing

3.1 Rationale for Preprocessing

Microscopic images captured in field conditions exhibit substantial variability in illumination, contrast, background texture, and sensor noise Meijering et al. (2016). Preprocessing was therefore essential to enhance diagnostically relevant features while suppressing irrelevant information Gonzalez and Woods (2018).

3.2 Image Resizing and Normalization

All images were resized to a fixed spatial resolution of 224×224 pixels, consistent with standard CNN input requirements He et al. (2016), Simonyan and Zisserman (2015). Pixel intensities were normalized using ImageNet statistics, a widely adopted practice that improves convergence and numerical stability Deng et al. (2009).

3.2.1 | Noise Reduction

Noise-handling strategies included consideration of Gaussian and median filtering to suppress sensor noise while preserving diagnostically important morphological features Lim (1990). Parameters were selected empirically to avoid degradation of spore structure.

3.3 Contrast Enhancement

Contrast enhancement methods such as histogram equalization and adaptive contrast techniques were employed to improve spore visibility under heterogeneous imaging conditions Pizer et al. (1987). This step was critical for distinguishing pebrine spores from non-pathogenic artifacts.

3.4 Data Augmentation

To improve model robustness, data augmentation techniques such as random cropping and flipping were applied during training Shorten and

Khoshgoftaar (2019). These operations simulated natural variability encountered in diagnostic practice, including slide orientation changes and microscope alignment differences.

4 Image Analysis

4.1 Theoretical Basis

Image analysis relied on **Convolutional Neural Networks (CNNs)**, which are well suited for hierarchical feature extraction from visual data. CNNs automatically learn spatial patterns and textures relevant to microscopic pathogen detection Litjens et al. (2017).

4.2 Model Architectures

Multiple CNN architectures were evaluated to compare classification accuracy, computational efficiency, and suitability for edge deployment Tan and Le (2020). Architectures included ResNet-50 He et al. (2016), EfficientNet variants Tan and Le (2020), and VGG-16 Simonyan and Zisserman (2015). Additionally, a microscopy-pretrained model (NASA MicroNet) was used to leverage domain-specific feature learning Stuckner et al. (2022).

4.3 Training Procedure

Models were trained using supervised learning with categorical cross-entropy loss Goodfellow et al. (2016). Optimization employed adaptive gradient-based methods, including Adam Kingma and Ba (2017) and stochastic gradient descent with momentum Sutskever et al. (2013). Hyperparameters were optimized using Bayesian optimization via Optuna Akiba et al. (2019).

4.4 Feature Learning and Representation

CNN layers progressively learned low-level features such as edges and textures, followed by higher-level representations corresponding to spore morphology and spatial organization Zeiler and Fergus (2014).

4.5 Classification and Decision Logic

Final network layers produced probabilistic outputs for class membership. Decision thresholds were selected to balance sensitivity and specificity, prioritizing reduction of false negatives due to the high risk associated with missed pebrine infections Fawcett (2006).

4.6 Evaluation Metrics

Model performance was evaluated using standard diagnostic metrics including accuracy, precision, recall, F1-score, and confusion matrices Powers (2020). Inference latency was measured to assess real-time applicability.

4.7 Edge Deployment and Optimization

Models were exported to ONNX format and evaluated using ONNX Runtime for efficient CPU inference ONNX developers (2026), ONNX Runtime developers (2018). Real-time camera-based inference was implemented using OpenCV OpenCV developers (2026). TorchScript Lite models were generated for deployment on resource-constrained devices Paszke et al. (2019).

4.8 Non-CNN Feature Engineering and Classical Classification Pipeline

To complement the CNN-based workflow, a parallel non-CNN pipeline was designed using handcrafted feature extraction and lightweight machine-learning classifiers. In contrast to CNNs, which learn hierarchical representations directly from pixel data, the non-CNN approach explicitly models diagnostically relevant texture and morphology from haemolymph microscopy images. This design was included to improve interpretability, reduce computational demand, and support deployment on low-power devices used in field sericulture settings Haralick et al. (1973), Ojala et al. (2002), Duda and Hart (1972), Cortes and Vapnik (1995), Breiman (2001).

4.8.1 | Design Rationale and System Positioning

The primary role of the non-CNN branch was not to replace deep learning, but to provide a biologically grounded and computationally efficient diagnostic pathway. In pebrine microscopy, several useful cues can be described explicitly: infected samples often exhibit altered texture, irregular local intensity patterns, and spore-like rounded structures. These cues can be translated into measurable descriptors and evaluated using classical classifiers. Such a framework is advantageous when dataset size is limited, when deployment must occur on edge/mobile hardware, or when interpretability is important for expert validation and troubleshooting Litjens et al. (2017), Shi et al. (2016), Satyanarayanan (2017).

4.8.2 | Preprocessing for Stable Feature Extraction

Because microscopy images were acquired under heterogeneous field and laboratory conditions, preprocessing was necessary to stabilize the extracted descriptors. Images were first converted to grayscale to standardize intensity representation. Noise reduction was then applied to

suppress high-frequency artifacts arising from sensor noise, slide impurities, and biological debris. Finally, contrast normalization was used to improve the visibility of diagnostically important structures, especially potential spores and infected regions. These steps reduced sensitivity to acquisition variability and improved the consistency of downstream feature computation Gonzalez and Woods (2018), Meijering et al. (2016).

4.8.3 | Texture-Based Feature Extraction Using GLCM

Global texture characteristics were quantified using Gray Level Co-occurrence Matrix (GLCM) features, following the framework introduced by Haralick et al. Haralick et al. (1973). Rather than analyzing pixels independently, GLCM models the spatial co-occurrence of gray levels at specified offsets and orientations. This makes it suitable for distinguishing infected haemolymph, which often shows dense granular appearance, irregular intensity transitions, and clustered spatial organization, from relatively smooth and homogeneous healthy samples.

From each image, second-order texture statistics were derived from the co-occurrence matrices, including contrast, correlation, energy, and homogeneity. These descriptors were selected because they quantify granularity, structural disorder, and local uniformity in a compact form. In the present application, they provided an interpretable description of infection-associated texture changes while remaining computationally inexpensive Haralick et al. (1973).

4.8.4 | Micro-Texture Analysis Using Local Binary Patterns

To capture fine-scale local irregularities that may not be fully represented by GLCM descriptors, Local Binary Patterns (LBP) were additionally extracted Ojala et al. (2002). LBP encodes the local neighborhood of each pixel by thresholding surrounding intensities relative to the center pixel, thereby generating a compact representation of micro-texture. This is particularly relevant in microscopy images where early or subtle infection may appear as small-scale local disturbances before more obvious global texture changes emerge.

The use of LBP was motivated by three practical considerations. First, pebrine-associated structures may introduce fine micro-patterns that are not always well described by broader texture statistics. Second, LBP histograms provide strong discriminative information with low computational cost. Third, rotation-invariant and uniform-pattern formulations improve robustness to orientation differences and minor acquisition variation, which are common in field microscopy Ojala et al. (2002). Thus, LBP complemented GLCM by refining discrimination at the local structural level.

4.8.5 | Morphological Feature Extraction Using the Hough Transform

A biologically important cue in pebrine diagnosis is the presence of spore-like rounded or near-elliptical structures. To incorporate this cue explicitly, Hough-transform-based detection was used as a morphology-oriented feature extractor Duda and Hart (1972). While CNNs may learn

such patterns implicitly, Hough-based analysis makes the geometric assumption transparent and directly testable.

The transform was used to identify candidate circular structures within a constrained radius range relevant to the expected spore scale. From this process, morphology-related descriptors such as candidate count, approximate density, and detection consistency were derived. These features were useful for distinguishing probable spores from irregular debris, fragmented tissue, or imaging artifacts. By introducing explicit geometric constraints, the method strengthened biological alignment and reduced dependence on purely statistical appearance differences Duda and Hart (1972).

4.8.6 | Feature Fusion Strategy

No single handcrafted descriptor was sufficient to capture the full variability of microscopic haemolymph samples. Therefore, the final non-CNN representation was constructed by concatenating complementary feature groups: GLCM-based texture statistics, LBP-based micro-texture histograms, and Hough-derived morphological descriptors. This fusion strategy generated a multi-dimensional representation containing global texture, local pattern, and structural shape information simultaneously.

The rationale for feature fusion was twofold. First, infection-related variability is multi-scale: some changes are diffuse and textural, whereas others are localized or morphological. Second, combining descriptors improves robustness against nuisance variation such as lighting changes, clutter, and sample-to-sample biological heterogeneity. In practice, fused features provided a more stable and informative representation than any single descriptor family alone Haralick et al. (1973), Ojala et al. (2002), Duda and Hart (1972).

4.8.7 | Classical Classification Framework

The fused handcrafted features were evaluated using a set of lightweight classifiers appropriate for small to moderate datasets. Support Vector Machines (SVMs) were included because they are effective in high-dimensional feature spaces and can provide strong margins between classes Cortes and Vapnik (1995). Random Forests were used because they are robust to noisy predictors, can model nonlinear feature interactions, and often perform well without extensive feature scaling Breiman (2001). Logistic Regression was included as an interpretable linear baseline that provides probabilistic outputs, and k-Nearest Neighbors (k-NN) served as a simple similarity-based reference model grounded in local neighborhood structure Cover and Hart (2006).

These classifiers were selected not only for predictive utility, but also for deployment practicality. Compared with deep CNNs, they require substantially fewer computational resources at inference time and are easier to inspect diagnostically. This makes them suitable for edge-assisted screening workflows in rural grainage environments Shi et al. (2016), Satyanarayanan (2017).

4.8.8 | Robustness, Parameter Sensitivity, and Error Analysis

Additional analysis was performed to understand the stability of the handcrafted pipeline. For GLCM, descriptor behavior depends on the chosen pixel distance and orientation; therefore, multiple directions were considered to reduce orientation bias and improve robustness to rotation. For LBP, neighborhood radius and pattern encoding influence sensitivity to fine structures; smaller radii were emphasized to preserve local detail relevant to spore-associated irregularities. For Hough-based detection, radius limits and detection thresholds were tuned conservatively because overly broad parameter ranges increased false positives from debris.

Error analysis further indicated that false positives were commonly associated with dense background artifacts or biological debris that resembled spore-like structures, while false negatives were more likely when spores were sparse, visually obscured, or weakly contrasted against the background. These observations support the use of a multi-feature and multi-stage diagnostic strategy rather than reliance on a single visual cue alone. The analysis also reinforced the importance of standardized imaging and careful calibration during deployment Gonzalez and Woods (2018), Meijering et al. (2016).

4.8.9 | CNN Versus Non-CNN Perspective

The CNN and non-CNN branches were treated as complementary rather than competing approaches. CNNs are advantageous for automatic hierarchical representation learning and can achieve high performance when sufficient data and compute are available Litjens et al. (2017). However, the handcrafted pipeline provides explicit biological interpretability, lower inference cost, and easier debugging. In field deployment scenarios where transparency, repeatability, and resource efficiency are essential, these properties are valuable.

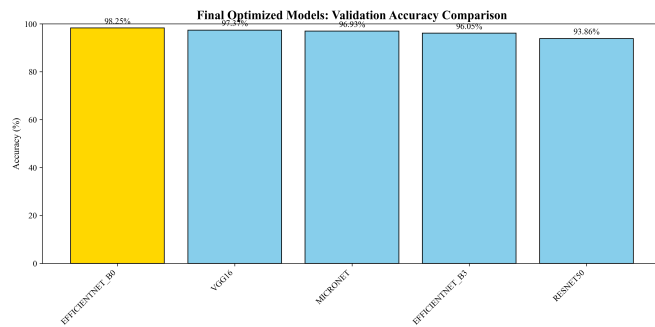
Accordingly, the non-CNN branch serves three practical purposes in the present study: (1) it acts as an interpretable baseline against which CNN behavior can be contextualized; (2) it provides a lower-cost deployment option for constrained hardware; and (3) it offers biologically meaningful intermediate signals that can help validate or challenge automated predictions. This combined design strengthens the overall diagnostic framework and supports future hybrid systems in which explicit handcrafted cues are integrated with learned representations.

4.9 Scientific Rigor and Reproducibility

All preprocessing steps, parameters, and evaluation protocols were documented systematically to ensure reproducibility Sandve et al. (2013). The experimental design supports replication across sericulture regions and extension to other microscopy-based animal health diagnostics.

TABLE 1 Conceptual comparison between the CNN and non-CNN pipelines used in the SeriScope framework.

Aspect	CNN Pipeline	Non-CNN Pipeline
Feature learning	Automatic hierarchical feature learning	Manually engineered texture and morphology features
Interpretability	Lower, because internal features are learned implicitly	Higher, because each descriptor has explicit visual meaning
Data requirement	Typically higher	Moderate and often more practical for limited datasets
Compute requirement	Higher during training and deployment	Lower and suitable for lightweight inference
Deployment suitability	Strong when optimized, but more hardware demanding	Well suited for edge and mobile screening workflows
Biological transparency	Indirect	Direct, through texture and shape-based descriptors

**FIGURE 5** Test accuracy comparison across evaluated CNN architectures.

5 Results and Performance Evaluation

This section reports classification performance, model efficiency, training behavior, explainability, and deployment-oriented evaluation. All metrics follow standard definitions for multi-class classification and diagnostic evaluation Powers (2020), Fawcett (2006). Inference latency is reported as mean milliseconds per image under CPU execution to reflect edge-style constraints.

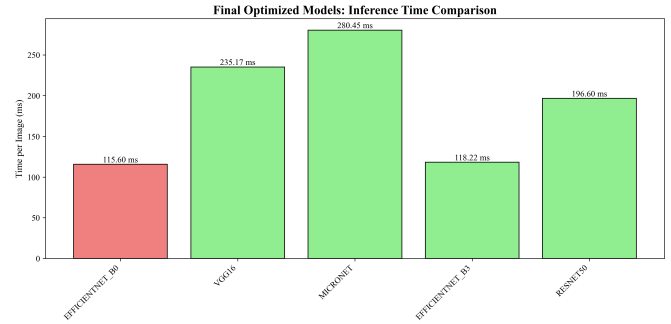
5.1 Overall Model Performance

5.1.1 | Accuracy and Inference Time Comparison

Figure 5 compares test accuracy across the evaluated architectures, while Figure 6 compares inference latency.

5.1.2 | Best-Observed Performance Across Runs

Because experiments were conducted across multiple runs (with different optimization settings and runtime contexts), we report the *best*

**FIGURE 6** Inference latency (ms/image) comparison across evaluated CNN architectures under CPU execution.**TABLE 2** Best observed PyTorch (PTH) test accuracy per architecture across multiple runs.

Model	Best Accuracy (%)
VGG16	98.68
EfficientNet-B0	98.25
MicroNet	96.93
EfficientNet-B3	96.05
ResNet-50	93.86

TABLE 3 ONNX Runtime evaluation summary.

Model (ONNX)	Accuracy (%)	Latency (ms/img)
MicroNet	96.32	16.53
VGG16	96.84	47.47
EfficientNet-B0	96.67	14.99
EfficientNet-B3	95.70	19.22
ResNet-50	93.07	18.88

observed test accuracy achieved by each architecture from the collected logs. This provides an upper-bound view of achievable performance under the evaluated pipeline. For deployment decisions, accuracy must be considered jointly with inference latency under the intended runtime environment.

5.1.3 | ONNX Export Evaluation

To validate deployment feasibility, selected models were exported to ONNX and evaluated using ONNX Runtime ONNX developers (2026), ONNX Runtime developers (2018). Because ONNX execution may differ from PyTorch due to operator implementations and optimization kernels, we report ONNX accuracy and latency separately from PyTorch results. Table 3 summarizes these outcomes.

5.1.4 | Best-Performing Model (PyTorch)

Across the compiled runs, the best observed PyTorch performance was obtained with VGG16, reaching a peak test accuracy of 98.68% (Table 2). The corresponding confusion matrix (Figure 8) indicates strong separability across the three classes (bacteria, non-pebrine, pebrine), with

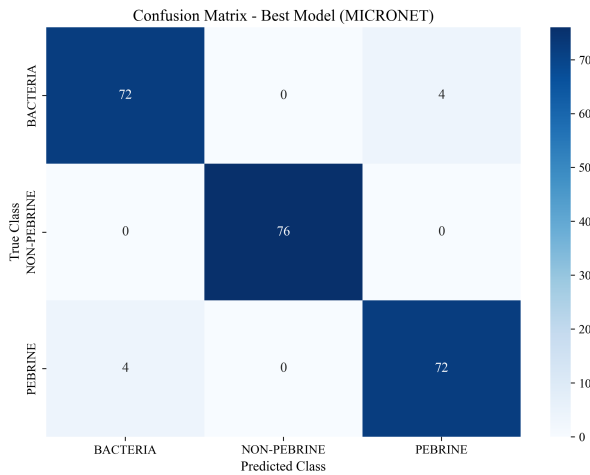


FIGURE 7 Confusion matrix for MicroNet.

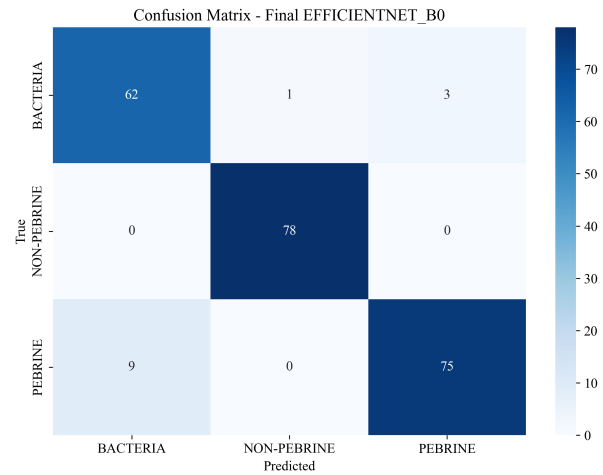


FIGURE 9 Confusion matrix for EfficientNet-B0.

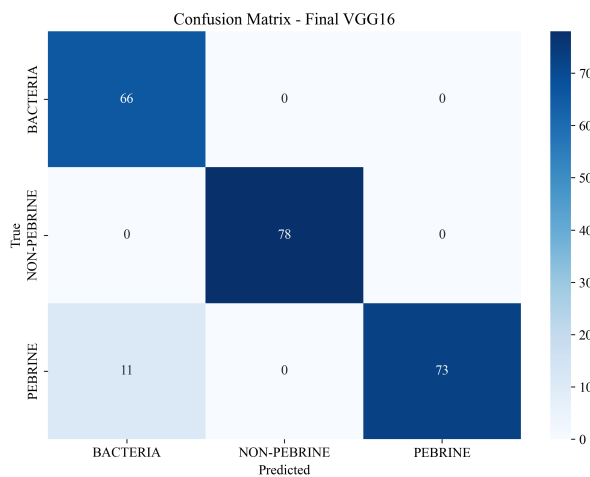


FIGURE 8 Confusion matrix for VGG16.

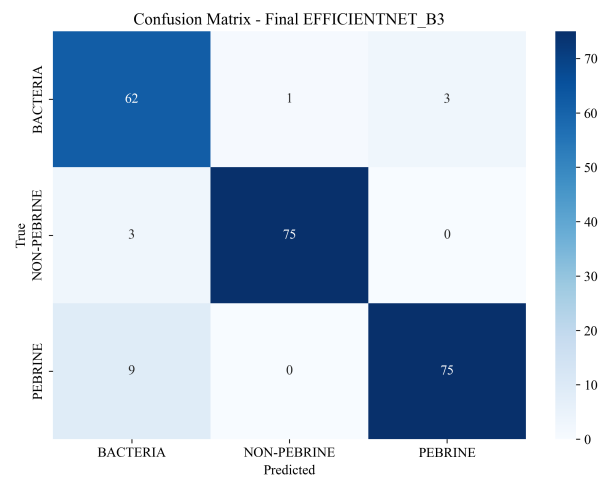


FIGURE 10 Confusion matrix for EfficientNet-B3.

remaining errors concentrated in visually ambiguous samples where artifact patterns resemble spore morphology.

5.2 Confusion Matrix Analysis

Confusion matrices for each evaluated architecture are provided in Figures 7–11. These illustrate class-wise error patterns and highlight misclassifications between pebrine spores and visually similar artifacts.

5.3 Training Dynamics

Training curves for each architecture are shown in Figures 12–16. These plots summarize convergence behavior and generalization trends, consistent with validation monitoring and early stopping practices Kohavi (1995).

5.4 Hyperparameter Optimization Behavior

Hyperparameter optimization was conducted using Optuna Akiba et al. (2019). Figure 17 shows the best-value trajectory over trials, illustrating systematic selection of learning rate, batch size, and regularization settings.

5.5 Explainability via Grad-CAM

To verify that model predictions are driven by biologically relevant visual cues, Grad-CAM visualizations were generated for representative test images. Figure 18 shows activation maps for each class, indicating regions most influential to the final decision Zeiler and Fergus (2014).

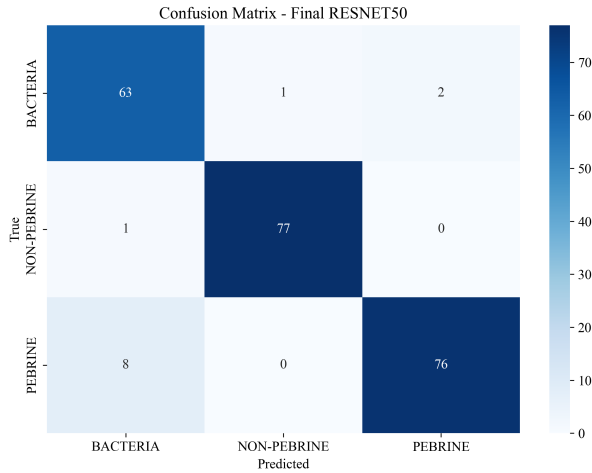


FIGURE 11 Confusion matrix for ResNet-50.

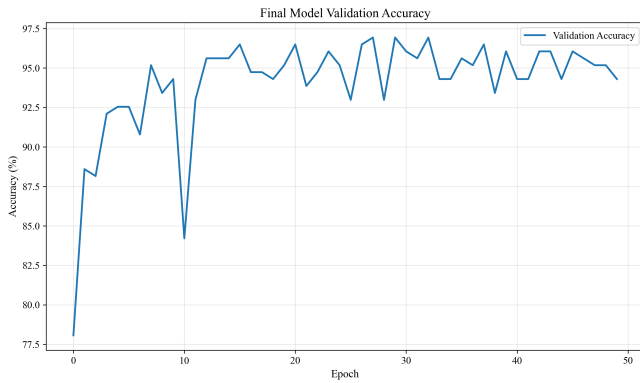


FIGURE 12 Training/validation curves for MicroNet.

5.6 Deployment Demonstration and User Interface

Figure 19 illustrates the SeriScope user interface used during real-time inference. The interface displays the live microscopy feed along with the predicted class label and confidence score, enabling rapid and objective decision-making at the field level.

5.7 Results: Non-CNN Feature Engineering Pipeline

The non-CNN pipeline was evaluated to quantify its effectiveness in distinguishing pebrine-infected and healthy haemolymph samples using handcrafted features and classical classifiers. The analysis focuses on three aspects: (i) discriminative capability of individual feature sets, (ii) improvements through feature fusion, and (iii) computational efficiency relative to CNN-based approaches.

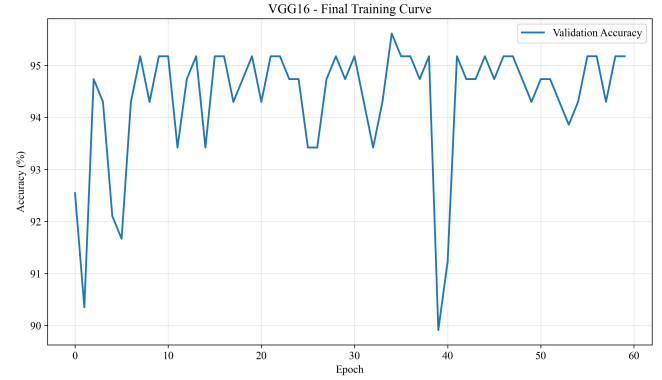


FIGURE 13 Training/validation curves for VGG16.

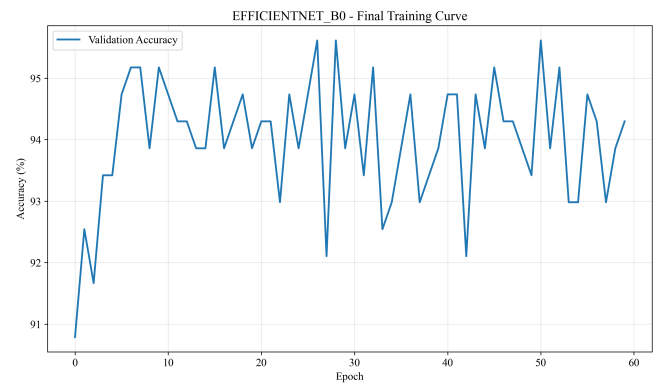


FIGURE 14 Training/validation curves for EfficientNet-B0.

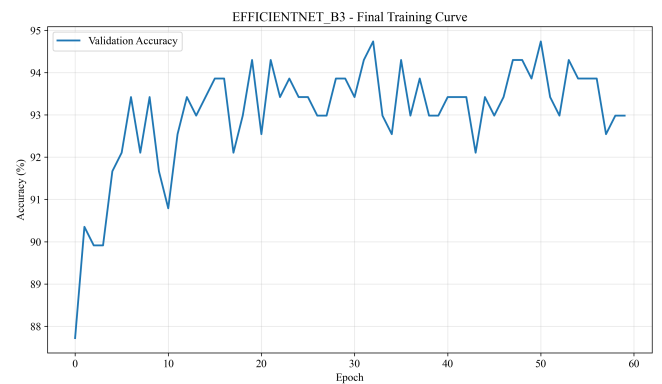


FIGURE 15 Training/validation curves for EfficientNet-B3.

5.7.1 | Performance of Individual Feature Sets

Each feature family was first evaluated independently to understand its contribution to classification performance. GLCM-based features demonstrated stable and consistent performance due to their ability to capture global texture variations associated with infection. Infected samples, characterized by granular and irregular intensity distributions,

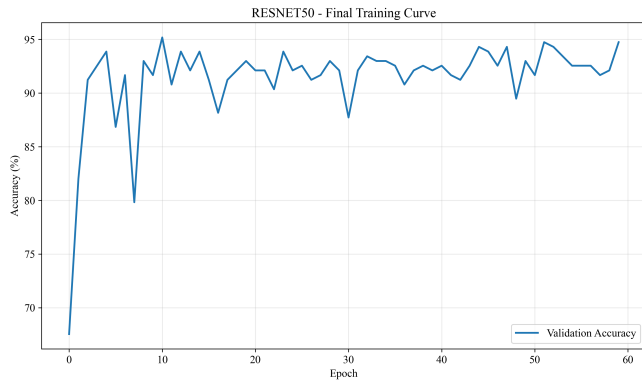


FIGURE 16 Training/validation curves for ResNet-50.

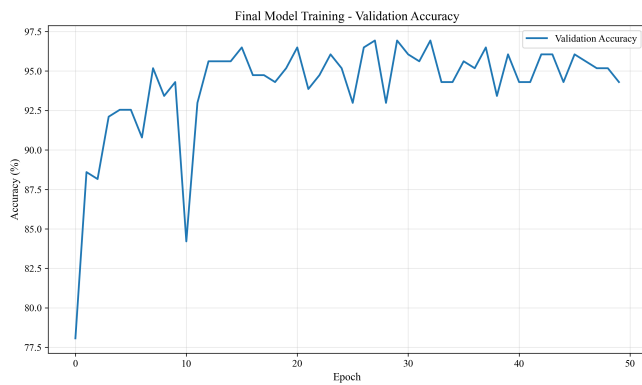


FIGURE 17 Optuna optimization curve: validation performance across trials Akiba et al. (2019).

were effectively differentiated from smoother healthy samples using second-order texture statistics.

LBP features showed improved sensitivity to fine-scale micro-texture variations. This was particularly important in early or low-density infection cases, where global texture differences were less pronounced but local irregularities were still detectable. However, LBP alone exhibited moderate variability under noisy conditions due to its dependence on local intensity relationships.

Hough-based morphological features provided biologically meaningful cues by detecting circular or near-circular structures consistent with spore-like formations. While this feature set aligned strongly with domain knowledge, its standalone performance was limited by susceptibility to false detections caused by debris, overlapping structures, and imaging artifacts.

Overall, while each feature set contributed useful information, none individually captured the full complexity of haemolymph microscopy patterns.

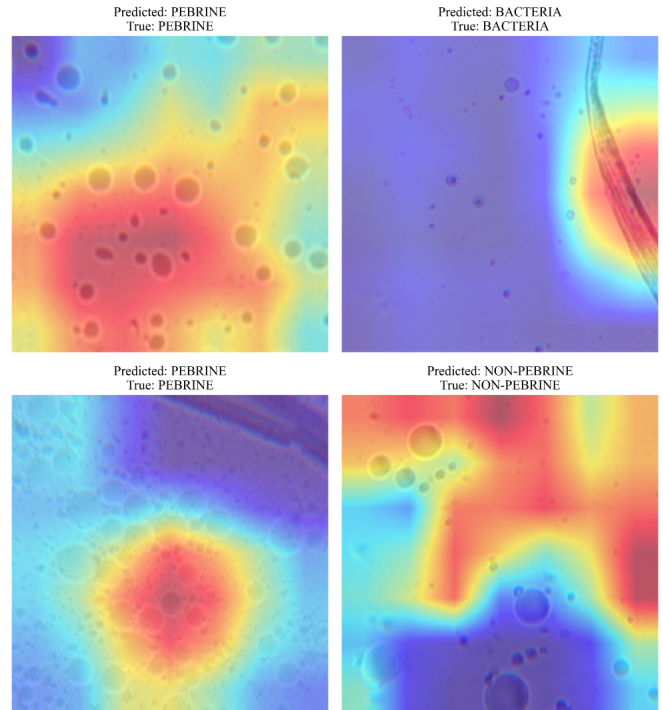


FIGURE 18 Grad-CAM examples (healthy/non-pebrine, bacteria, pebrine) showing model attention regions on microscopy images Zeiler and Fergus (2014).

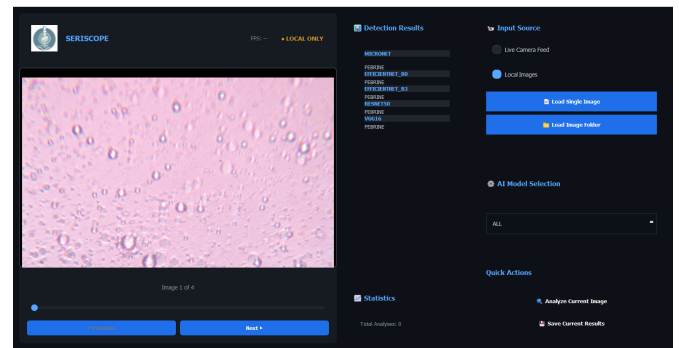


FIGURE 19 SeriScope user interface during real-time inference, showing the live microscopy input, predicted class label, and confidence score.

5.7.2 | Effect of Feature Fusion

The combination of complementary feature sets resulted in a significant improvement in classification performance. Specifically, the integration of GLCM and LBP features produced a more robust representation by jointly encoding global texture and local micro-patterns. This combination reduced misclassification in cases where either descriptor alone was insufficient.

The inclusion of Hough-derived morphological features further enhanced discrimination in samples containing clearly defined spore-like

structures. Although Hough features were less reliable in isolation, their contribution within the fused representation improved sensitivity to biologically relevant shapes.

The best-performing configuration was obtained using the combined feature vector (GLCM + LBP + Hough), which consistently outperformed all individual feature sets. This confirms that pebrine-related visual signatures are multi-scale in nature and require integrated representation across texture and morphology domains.

5.7.3 | Classifier Comparison

Among the evaluated classifiers, Support Vector Machines (SVM) achieved the most consistent performance across feature sets due to their ability to handle high-dimensional feature spaces and nonlinear class boundaries Cortes and Vapnik (1995). Random Forests also performed robustly, particularly in scenarios with noisy or partially redundant features, owing to their ensemble-based structure Breiman (2001).

Logistic Regression provided a stable but comparatively lower baseline, reflecting the limitations of linear decision boundaries for complex microscopy data. The k-Nearest Neighbors classifier showed sensitivity to feature scaling and dataset density, resulting in variable performance across experiments.

These observations suggest that margin-based and ensemble methods are better suited for the heterogeneous feature space generated by handcrafted descriptors.

5.7.4 | Computational Efficiency and Deployment Considerations

A key advantage of the non-CNN pipeline lies in its computational efficiency. Feature extraction and classification required significantly lower inference time compared to CNN-based models, making the approach suitable for edge deployment scenarios. The absence of deep network layers reduces both memory footprint and processing requirements, enabling faster predictions on resource-constrained hardware.

Latency analysis further indicated that feature extraction constitutes the primary computational cost within the pipeline, while classification remains relatively lightweight. Even with feature fusion, the total inference time remained substantially lower than CNN-based approaches, supporting real-time or near-real-time diagnostic applications in field conditions Shi et al. (2016), Satyanarayanan (2017).

5.7.5 | Error Analysis

Misclassifications were analyzed to identify limitations and guide future improvements. False positives were most commonly associated with dense background artifacts or biological debris that resembled infection-like texture or morphology. False negatives occurred in cases where spore presence was sparse, poorly contrasted, or partially occluded.

These findings highlight the importance of standardized sample preparation and imaging conditions, as well as the potential benefit of integrating confidence scoring or hybrid CNN-non-CNN systems for improved robustness.

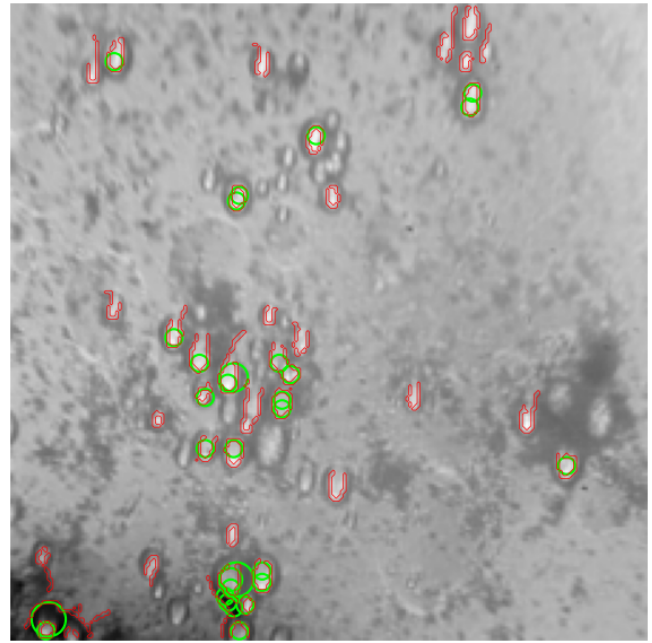


FIGURE 20 Representative Hough-circle detection showing candidate spore-like circular structures identified in a microscopy image.

5.7.6 | Comparison with CNN Pipeline

While CNN models generally achieved higher peak accuracy due to their ability to learn complex hierarchical representations Litjens et al. (2017), the non-CNN pipeline demonstrated competitive performance when feature fusion was applied. More importantly, it provided superior interpretability and significantly lower computational requirements.

This trade-off suggests that the non-CNN approach is particularly valuable for deployment in constrained environments, while CNN models may serve as higher-accuracy reference systems when computational resources permit. The combined use of both approaches forms a complementary diagnostic framework, balancing performance, efficiency, and explainability.

6 DISCUSSION

6.1 Future Improvements: Fluorescence and Molecular Diagnostics

While the current computer-vision-based pipeline demonstrates strong performance, further improvements are being explored to enhance biological specificity and reduce ambiguity.

6.1.1 | Fluorescence Microscopy Integration

Fluorescence microscopy is being evaluated as a complementary imaging modality to improve signal-to-noise ratio and enable clearer identification of *Nosema bombycis* spores. As noted in the mentorship findings

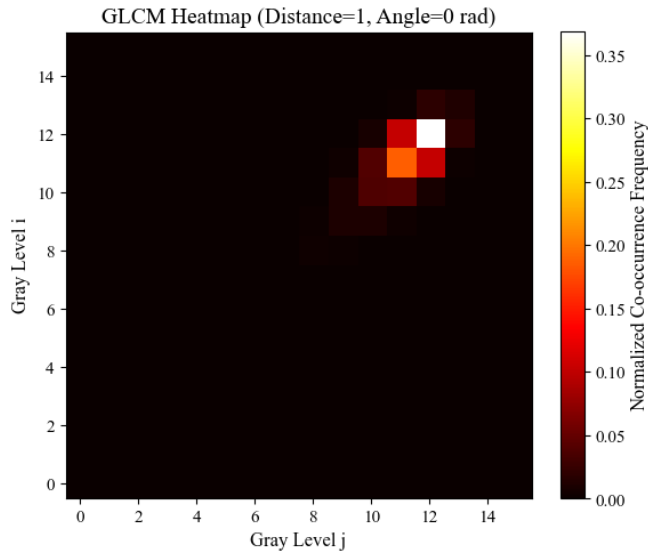


FIGURE 21 Gray Level Co-occurrence Matrix (GLCM) heatmap illustrating second-order texture relationships used for handcrafted feature extraction.

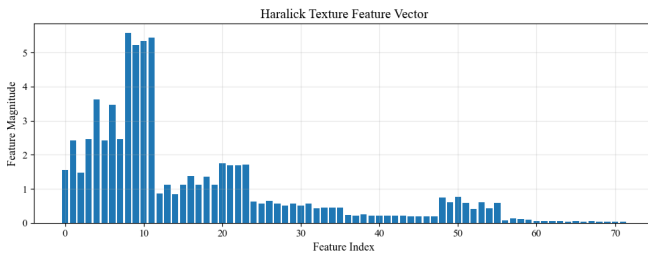


FIGURE 22 Haralick texture feature vector extracted from a representative microscopy image for quantitative texture characterization.

(pages 11–12), the use of Calcofluor White, a chitin-binding fluorescent dye, allows selective visualization of microsporidian spores under UV excitation.

This approach offers two key advantages:

- Improved boundary definition of spores, enabling more accurate morphological detection (e.g., Hough-based counting)
- Reduction of background noise and ambiguity caused by debris or host tissue

To maintain cost-effectiveness, an external oblique UV illumination setup is being explored as an alternative to expensive epifluorescence systems.

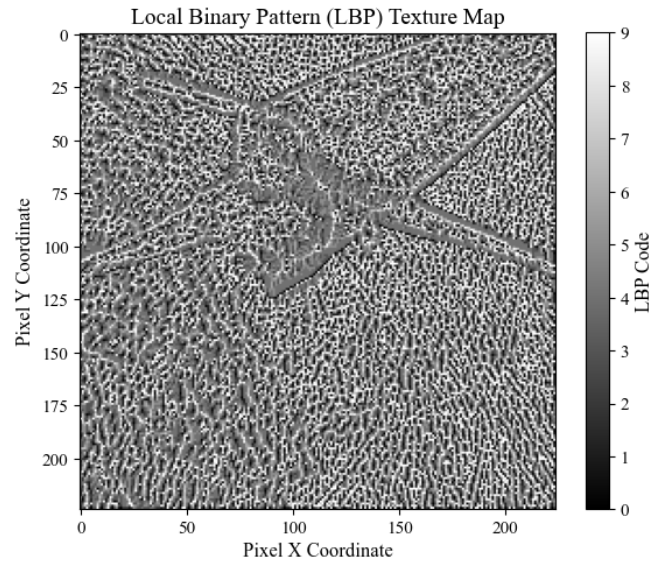


FIGURE 23 Local Binary Pattern (LBP) map showing fine-scale local texture variations used to capture microstructural irregularities.

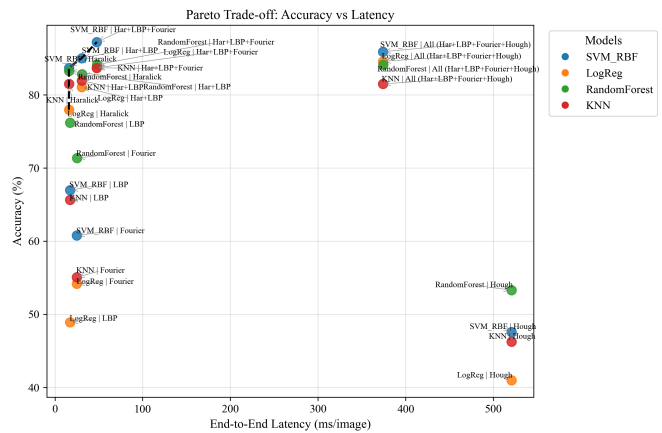


FIGURE 24 Pareto analysis of non-CNN model-feature configurations showing the trade-off between classification accuracy and end-to-end latency.

6.1.2 | DNA-Based Molecular Confirmation

In addition to imaging-based approaches, molecular diagnostic techniques are being investigated to provide confirmatory evidence of infection. These include:

- PCR/qPCR-based detection of *Nosema bombycis*
- Loop-mediated isothermal amplification (LAMP) for field-deployable testing
- Emerging CRISPR-based detection systems for rapid and sensitive identification

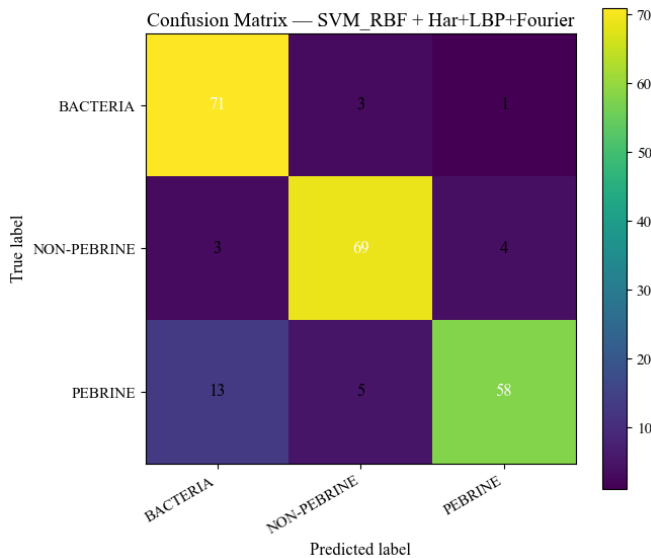


FIGURE 25 Confusion matrix of the best-performing non-CNN configuration, summarizing class-wise prediction performance.

Recent studies indicate that LAMP assays can detect *N. bombycis* with high specificity within approximately 50 minutes, while PCR-based methods offer high sensitivity even in early-stage infection.

The most practical deployment strategy is a hybrid workflow:

Computer vision-based screening → *Targeted molecular confirmation*

This approach combines scalability and speed of AI-based microscopy with the biological certainty of molecular diagnostics, enabling robust and field-deployable disease detection.

6.1.3 | Summary of Improvements

Overall, the post-fair refinements significantly strengthen the system by:

- Improving computational efficiency through non-CNN methods
- Correcting biological misclassification of ambiguous structures
- Expanding dataset size and validation robustness
- Introducing multi-modal diagnostic pathways (optical + molecular)

These advancements enhance both the scientific validity and practical applicability of the SeriScope framework.

7 CONCLUSIONS

This research shows how to combine animal science, microscopy, and edge artificial intelligence for early disease diagnosis in sericulture World Organisation for Animal Health (2024), Hallou et al. (2021), Shi et al. (2016). Image analysis using convolutional neural networks

supports reliable classification in microscopy-driven diagnostics Litjens et al. (2017). The emphasis on reducing false negatives is consistent with diagnostic-risk trade-offs in classification settings Fawcett (2006), Powers (2020). A key contribution is offline edge deployment, which reduces latency and infrastructure dependence in resource-limited environments Shi et al. (2016), Satyanarayanan (2017). These field-deployable diagnostic capabilities support productivity and rural innovation objectives aligned with SDG 8 and SDG 9 United Nations (2015).

8 DATA AVAILABILITY STATEMENT

The datasets generated and analyzed during this study are available from the corresponding author upon reasonable request Sandve et al. (2013). Data sharing depends on institutional and field-level collaboration agreements.

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